Statistics for Empirical CAPM Testing

A from-scratch companion to Pankaj Chaudhary's "Testing of CAPM in Indian Context", with worked examples from YOUR own Nifty 500 results

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This guide assumes **no prior statistics**. It builds from the basics (average, spread, bell curve) up to the exact t-test / p-value machinery used to test CAPM — and then works through the actual numbers your own Python analysis produced, so the theory and your output line up. The t-test sections follow the clear, step-by-step style of the DATAtab "t-Test – Full Course" video.

The one-sentence goal. CAPM says a stock's reward (its return above a safe asset) should depend only on **beta** — how much it swings with the market. Empirical testing asks, with data + statistics: *is that actually true?*

Part 0 — The building blocks (start here)

0.1 Data, sample, and population

A **variable** is anything we measure (a stock's monthly return). Each measurement is an **observation**. A **population** is every value that could ever occur; a **sample** is the slice we actually have (say 60 months). Statistics judges the whole population from a limited sample, so there is always some uncertainty — which is exactly what t-values and p-values measure.

Formula

```
population = everything (unknown) sample = what we observe (60 or 120 monthly returns)
```

Simple example

You cannot taste a whole pot of soup; you stir and taste one spoon (sample) and infer the pot (population).

In the CAPM paper

Your sample is 2016–2025 monthly returns of Nifty 500 stocks; the paper's is 2006–2015.

0.2 Mean (the average)

The **mean** is the balancing point: add the values and divide by how many. It answers "what is typical?".

Formula

```
mean = (sum of values) ÷ (count)
```

Simple example

Returns 2%, −1%, 3% → mean = 4 ÷ 3 = 1.33% per month.

In the CAPM paper

"Mean Excess Return" per portfolio (your Sheet 4; the paper's "Average Return" in Table 2) is exactly this.

0.3 Variance and standard deviation (spread = risk)

Variance measures spread: average of (value − mean)², squared so plus and minus don't cancel. **Standard deviation** = √variance, back in normal units. Large spread = risky / unpredictable.

Formula

```
variance = average of (value − mean)² standard deviation = √variance
```

Simple example

Two stocks average 1%/month. A swings −10% to +12%; B stays 0.5%–1.5%. Same mean, but A's variance is far bigger — A is riskier.

In the CAPM paper

Standard deviation of returns is the standard measure of total risk; CAPM then splits it into two parts (see 3.5).

0.4 The normal distribution (bell curve)

The **normal distribution** is the symmetric "bell": most values cluster near the mean; extremes are rare. It matters because sample averages and regression estimates follow a bell shape, which lets us attach probabilities (p-values) to results.

Formula

```
≈ 68% of values lie within 1 standard deviation of the mean; ≈ 95% within 2.
```

Simple example

Adult heights form a bell; monthly market returns look roughly bell-shaped too.

In the CAPM paper

The "95% within 2 SD" fact is the origin of the famous "|t| > 2 means significant" rule used in every results table.

0.5 Correlation

Correlation measures whether two variables move together, from −1 to +1. +1 = perfectly together, 0 = no linear link, −1 = opposite. It is the seed of beta and R^2.

Formula

```
near +1: move together | near 0: unrelated | near −1: move oppositely
```

Simple example

Ice-cream sales and temperature: high positive correlation.

In the CAPM paper

Beta is a cousin of correlation — how strongly a stock moves with the market, scaled by their swing sizes.

Part 1 — The logic of a statistical test

1.1 Signal vs luck

If high-beta portfolios earned more in your sample, two stories compete: a real **signal** (beta is truly rewarded) or mere **luck** (this decade happened to look that way). A test decides which is more believable.

Simple example

Flip a coin 10 times, get 7 heads. Biased (signal) or just luck for a fair coin? A test answers with a number.

In the CAPM paper

Every t-Statistic and Prob. in the paper's Tables 3–5 is separating real effects from chance.

1.2 Null (H0) and alternative (H1) hypotheses

The **null H0** is the cautious default: “no effect / value = 0.” The **alternative H1** is what you accept only if the evidence is strong. You assume H0 and check whether data force you to reject it. You never prove H1.

Formula

```
H0: coefficient = 0 H1: coefficient ≠ 0 (or > 0)
```

Simple example

Court: “innocent” (H0) is assumed; only strong evidence yields “guilty” (H1).

In the CAPM paper

For the premium: H0: $\lambda_1 = 0$ (beta earns nothing) vs H1: $\lambda_1 > 0$. For alpha: H0: $\alpha = 0$ (CAPM prices it right).

1.3 Significance level (α)

Before testing, pick the false-alarm risk you accept — the **significance level α**, usually 5% or 1%. If the evidence beats this bar, you reject H0 and call the result **significant**.

Formula

```
α = 0.05 → reject H0 only if a result this extreme would happen < 5% of the time by chance.
```

Simple example

A smoke alarm set too sensitive (big α) beeps at toast; stricter (α = 1%) beeps less but may miss a small fire.

In the CAPM paper

The paper marks * = significant at 5% and ** = at 1% — exactly these two α levels.

1.4 Type I and Type II errors

Type I (false alarm) = rejecting a true H0; its probability = α. **Type II** (missed detection) = failing to reject a false H0. **Power** = 1 − Type II probability = ability to catch a real effect.

Formula

```
Type I probability = α Power = 1 − Type II probability
```

Simple example

Medical test: calling a healthy person sick = Type I; calling a sick person healthy = Type II.

In the CAPM paper

Grouping stocks into 25–32 portfolios cuts noise and raises power, so a real premium is not missed.

1.5 One-tailed vs two-tailed

Two-tailed: “is it different from H0?” (either direction). **One-tailed**: “is it greater?” (or less). One-tailed puts all α on one side, so it needs a slightly smaller t in that direction.

Formula

```
two-tailed 5%: reject if |t| > ≈ 2.0 one-tailed 5%: reject if t > ≈ 1.7
```

Simple example

“Does a course change scores?” is two-tailed; “does it raise scores?” is one-tailed.

In the CAPM paper

“Is $\alpha \neq 0$?” is two-tailed. “Is $\lambda_1 > 0$?” is naturally one-tailed (theory predicts a positive sign).

Part 2 — The t-test, the t-table, and the p-value

2.1 Standard error (SE): the ± on an estimate

A number from a sample would differ with another sample. The **standard error** measures that wobble — the precision of an estimate. Smaller SE = sharper. It shrinks as the sample grows.

Formula

```
SE of an average = s ÷ √n (s = standard deviation, n = sample size)
```

Simple example

Average return 1.0% with SE 0.2% → true average is about 1.0% ± ~0.4%.

In the CAPM paper

Every β, α and λ has a hidden SE (your code's se = sqrt(diag(...))). The t-value is estimate ÷ SE.

2.2 The t-test — what it is and the 5 steps

A **t-test** checks whether a mean (or a coefficient) is really different from some value, using the **t-statistic** = how many standard errors the estimate sits from that value. There are three common flavours: **one-sample** (compare one mean to a fixed number), **independent two-sample** (compare two groups), and **paired** (before vs after). A regression coefficient's test is the one-sample kind, with the fixed number = 0.

The recipe (same every time)

1. State H0 and H1. **2.** Compute t = (estimate − H0 value) ÷ SE. **3.** Find degrees of freedom (df). **4.** Look up the critical t for your α and df (t-table), *or* get the p-value. **5.** Decide: if |t| > critical t (equivalently p < α), reject H0.

Formula (one-sample)

```
t = (sample mean − hypothesised value) ÷ (s ÷ √n) df = n − 1
```

Worked example (video-style)

A coaching centre claims the average test score is **70**. You sample **n = 16** students: sample mean = **74**, sample SD s = **8**.
Step 1 — H0: mean = 70; H1: mean ≠ 70.
Step 2 — SE = 8 ÷ √16 = 8 ÷ 4 = 2, so t = (74 − 70) ÷ 2 = **2.0**.
Step 3 — df = 16 − 1 = 15.
Step 4 — critical t (5%, two-tailed, df 15) = **2.131** (from the t-table).
Step 5 — |2.0| < 2.131, so we **fail to reject H0**: the score is not significantly different from 70 (the p-value ≈ 0.064, just above 0.05, agrees).

2.3 The critical t-value and the t-table

The **critical t-value** is the cut-off your t must beat to be significant. It depends on two things: the significance level **α** (how strict) and the **degrees of freedom df** (roughly the sample size minus the number of things you estimated). You read it from a **t-table**: find your df row, slide to your α column. Two patterns to notice: as df grows the critical value **shrinks** toward the normal-curve values (1.645, 1.96, 2.58); and stricter α (1% vs 5%) needs a **bigger** t.

df	α=0.10 (two-tail)	α=0.05 (two-tail)	α=0.01 (two-tail)
1	6.314	12.706	63.657
2	2.920	4.303	9.925
3	2.353	3.182	5.841
4	2.132	2.776	4.604
5	2.015	2.571	4.032
6	1.943	2.447	3.707
7	1.895	2.365	3.499
8	1.860	2.306	3.355
9	1.833	2.262	3.250
10	1.812	2.228	3.169
11	1.796	2.201	3.106
12	1.782	2.179	3.055
13	1.771	2.160	3.012
14	1.761	2.145	2.977
15	1.753	2.131	2.947
16	1.746	2.120	2.921
17	1.740	2.110	2.898
18	1.734	2.101	2.878
19	1.729	2.093	2.861
20	1.725	2.086	2.845
21	1.721	2.080	2.831
22	1.717	2.074	2.819
23	1.714	2.069	2.807
24	1.711	2.064	2.797
25	1.708	2.060	2.787
26	1.706	2.056	2.779
27	1.703	2.052	2.771
28	1.701	2.048	2.763
29	1.699	2.045	2.756
30	**1.697**	**2.042**	**2.750**
40	1.684	2.021	2.704
60	1.671	2.000	2.660
120	1.658	1.980	2.617
∞ (z)	1.645	1.960	2.576

The orange row (df = 30) is the one your CAPM second-pass test uses — critical value **2.042** at 5%. Your first-pass tests use df = 58, whose critical value is ≈ 2.00.

2.4 The p-value (the shortcut that skips the table)

The **p-value** is the probability of getting a t at least as extreme as yours *if H0 were true* — it is the shaded area in the tail(s) of the bell beyond your t. A far-out t leaves a tiny tail area = tiny p = strong evidence against H0. Decision: **p < α → reject H0**. It gives the same verdict as the t-table, without looking anything up, which is why software (and your Python) reports it directly.

Formula

```
p = tail area beyond your t. p < 0.05 → significant. Big |t| ⇔ small p (they always agree).
```

Simple example

In the score example, t = 2.0 with df 15 leaves ≈ 6.4% in the two tails → p ≈ 0.064 > 0.05 → not significant. If t had been 3.0, p ≈ 0.009 < 0.05 → significant.

In the CAPM paper

The paper's “Prob.” column IS the p-value: Table 3 shows Prob = 0 for λ_0 and λ_1 (highly significant); Table 4 shows Prob = 0.2573 for λ_2 (not significant).

2.5 z-test (the large-sample cousin)

The **z-test** uses the plain normal curve and applies when the sample is large or the true spread is known. For big samples t and z nearly coincide (that is the ∞ row of the t-table).

Formula

```
z = (estimate − H0 value) ÷ SE reject at 5% (two-tailed) if |z| > 1.96
```

Simple example

With thousands of observations a t-test and z-test give essentially the same answer.

In the CAPM paper

Your 60–120 monthly rows are modest, so t is correct; z is for very long histories.

2.6 Confidence interval (CI): the range version

A **confidence interval** is a plausible range for the true value: estimate ± (critical t) × SE. If the range **excludes** the H0 value (usually 0), the result is significant — the same verdict as the t-test, shown as a range instead of a single number.

Formula

```
95% CI = estimate ± (critical t) × SE
```

Simple example

Estimate 0.020, SE 0.004, critical t ≈ 2 → CI = 0.020 ± 0.008 = [0.012, 0.028]. Excludes 0 → significant.

In the CAPM paper

A 95% CI for λ_1 that stays above 0 supports CAPM. In the paper it sits below 0 — significant, but wrong sign.

2.7 F-test / F-statistic

A t-test checks one coefficient; an **F-test** checks several at once, or whether the whole model explains a meaningful share of variation. Large **F** (small F p-value) = the group of variables jointly matters.

Formula

```
F compares explained variation to leftover variation across several coefficients
together.
```

Simple example

Do β^2 AND residual variance *together* improve the model? An F-test answers, even if neither shines alone.

In the CAPM paper

The overall significance of each second-pass model is an F-test; the famous GRS test is an F-test that all portfolio alphas are jointly zero.

Part 3 — Regression and the two kinds of risk

3.1 Regression: the line of best fit

Regression draws the straight line best fitting a cloud of points, summarising how output Y depends on input X. It has a **slope** (rise in Y per +1 of X) and an **intercept** (Y when X = 0). These are the coefficients we t-test.

Formula

```
Y = intercept + slope × X + error (error = vertical gap from the line)
```

Simple example

Plot a stock's monthly return (Y) against the market's (X); the best-fit slope shows how strongly it follows the market.

In the CAPM paper

First pass: portfolio excess on market excess → slope = **beta**, intercept = **alpha**. Second pass: average returns on beta → slope = premium λ_1.

3.2 Residual and residual variance

A **residual** is what the line misses at a point: actual − predicted. **Residual variance** is how spread out those misses are. Small = tight fit, little unexplained.

Formula

```
residual = actual − predicted residual variance = average of (residual)²
```

Simple example

Actual 3%, line predicts 2.5% → residual 0.5%. The variance of all such misses is the unexplained part.

In the CAPM paper

A portfolio's residual variance = its **unsystematic (firm-specific) risk**. The paper's Table 2 lists it as “Variance of Residuals”; Table 5 tests whether it is wrongly rewarded.

3.3 R-squared (R^2)

R-squared is the fraction of Y's ups-and-downs the model explains, 0 to 1. 0.60 = 60% explained, 40% residual. For one input it equals correlation².

Formula

```
R² = 1 − (residual variation ÷ total variation) = correlation² (single input)
```

Simple example

If beta explains 46% of the differences in average returns across portfolios, $R^2 = 0.46$.

In the CAPM paper

Paper Table 3: second-pass $R^2 = 0.537$ (beta explains ~54% of the spread) — decent fit, but the premium's sign is wrong.

3.4 Adjusted R-squared

Plain R^2 can only rise when you add inputs, even useless ones. **Adjusted R^2** penalises extra inputs, so it rises only if a new variable genuinely helps. Use it to compare models with different numbers of predictors.

Formula

```
adjusted R² = 1 − (1 − R²)(n − 1) ÷ (n − k − 1) (n = points, k = inputs)
```

Simple example

Paper Table 3 (1 input): R^2 0.537, adjusted 0.517. Table 5 (3 inputs): R^2 0.596 but adjusted only 0.539 — extras barely help.

In the CAPM paper

Use adjusted R^2 to compare your second-pass Models 1, 2, 3 fairly (they use 1, 2, 3 predictors).

3.5 Systematic vs unsystematic risk (beta and alpha)

CAPM splits total risk in two. **Systematic (market) risk** moves with the whole market, cannot be diversified away, and is measured by **beta**. **Unsystematic (firm-specific) risk** is unique to a company and can be diversified away by holding many stocks. CAPM's core claim: investors are rewarded *only* for systematic risk.

Formula

```
total risk = systematic (beta, cannot diversify) + unsystematic (firm-specific, can diversify)
```

Simple example

A factory fire (one firm) is unsystematic — it washes out in a big portfolio. A rate hike hits everything — systematic.

In the CAPM paper

This is the paper's spine: test whether beta is rewarded (Table 3) and whether unsystematic risk is wrongly rewarded (Table 5).

Part 4 — Reading this paper's result tables

The paper uses a **two-pass** method. **First pass:** for each of 25 portfolios, regress its excess return on the market's; the slope is **beta**, the leftover scatter is **residual variance**. **Second pass:** take one number per portfolio (average return, beta, beta², residual variance) and run one cross-sectional regression to see what explains average returns. Then read λ_0, λ_1, λ_2, λ_3 with t-tests.

Table 3 — Is beta rewarded? (the main test)

Regresses average return on beta: Return = $\lambda_0 + \lambda_1 \cdot$beta. CAPM predicts $\lambda_0 = 0$ and $\lambda_1 > 0$ (and λ_1 should equal the market premium).

Formula

```
λ₀ = 1.299 (t = 6.40, Prob = 0) λ₁ = −0.915 (t = −5.17, Prob = 0) R² = 0.537
```

Simple example

Both highly significant ($|t| \gg 2$, $p = 0$). But λ_0 is not 0 (should be) and λ_1 is negative (should be positive) — so higher-beta portfolios earned *less*, the opposite of CAPM.

What it means

Verdict: beta matters statistically but points the wrong way — plain beta fails for India.

Table 4 — Is the relationship curved? (non-linearity)

Adds beta-squared: Return = λ_0 + λ_1·beta + λ_2·beta². A significant λ_2 would mean a curved (non-CAPM) relationship.

Formula

λ_2 = −0.4102 (t = −1.16, Prob = 0.2573)

Simple example

p = 0.26 > 0.05, so λ_2 is not significant — the relationship is a straight line, no curvature.

What it means

Verdict: the failure is not due to curvature; linearity holds.

Table 5 — Is firm-specific risk wrongly rewarded?

Adds residual variance: Return = λ_0 + λ_1·beta + λ_2·beta² + λ_3·(residual variance). CAPM insists λ_3 = 0.

Formula

λ_3 = −0.0241 (t = −1.30, Prob = 0.2091); only λ_0 is significant (t = 2.29)

Simple example

λ_3 is not significant (p = 0.21), so firm-specific risk is not priced in the full period; but λ_1 is insignificant here too — beta lost its power once other terms entered.

What it means

Overall: λ_0 stuck away from 0 and λ_1 never positive → **CAPM does not hold for India** (matching Gupta & Sehgal 1993, Ansari 2000).

Part 5 — Worked examples using YOUR real Nifty 500 numbers

Now the payoff: below are the actual numbers your Python analysis produced, run through the exact 5-step t-test recipe from Part 2. This is where the theory and your Sheet 5 / first-pass output connect.

5.1 The CAPM premium λ_1 (second pass — the headline test)

Step	What we do	Result (real numbers)
1. Hypotheses	H0: $\lambda_1 = 0$ (beta earns nothing). H1: $\lambda_1 \neq 0$.	
2. Estimate & SE	from the cross-section regression across 32 portfolios	$\lambda_1 = 0.02041$, SE = 0.00406
3. t-value	t = estimate ÷ SE = 0.02041 ÷ 0.00406	t = 5.03
4a. df	df = n − k = 32 portfolios − 2 coefficients	df = 30
4b. critical t	t-table, 5% two-tailed, df 30 (orange row in 2.3)	2.042
4c. p-value	tail area beyond t = 5.03 (from software)	p = 0.0000212
5. Decide	|5.03| > 2.042 and p < 0.05	REJECT H0 → significant

Meaning: beta earns a positive, highly significant premium in 2016–2025 — higher-beta portfolios really did earn more. And $\lambda_1 = 0.0204$ sits above the average market excess return (0.0076), so beta was, if anything, over-rewarded.

5.2 The intercept λ_0 (second pass — should be zero under CAPM)

Step	What we do	Result (real numbers)
1. Hypotheses	H0: $\lambda_0 = 0$ (CAPM prices things right). H1: $\lambda_0 \neq 0$.	
2. Estimate & SE	same regression	$\lambda_0 = -0.00849$, SE = 0.00480
3. t-value	t = −0.00849 ÷ 0.00480	t = −1.77
4. df / critical / p	df = 30; critical = 2.042; p = 0.087	|1.77| < 2.042
5. Decide	|t| below the cut-off, p > 0.05	FAIL to reject H0

Meaning: the intercept is NOT significantly different from 0 — which is exactly what CAPM wants. Notice the contrast with the paper, where λ_0 WAS significantly non-zero (t = 6.40). Your decade behaves more like CAPM here.

5.3 A first-pass beta (Portfolio P1, 60 months)

Step	What we do	Result (real numbers)
1. Hypotheses	H0: $\beta = 0$ (P1 doesn't move with the market). H1: $\beta \neq 0$.	
2. Estimate & SE	time-series regression of P1 excess on market excess	$\beta = 1.3744$, SE = 0.1922
3. t-value	t = 1.3744 ÷ 0.1922	t = 7.15
4. df / critical / p	df = 60 − 2 = 58; critical ≈ 2.00; p ≈ 0.0000000016	|7.15| ≫ 2.00
5. Decide	far past the cut-off	REJECT H0 → significant

Meaning: P1 clearly moves with the market (beta ≈ 1.37, a high-beta basket). Almost every portfolio's beta is strongly significant — betas are measured reliably.

5.4 A first-pass alpha (Portfolio P1 — should be zero under CAPM)

Step	What we do	Result (real numbers)
1. Hypotheses	H0: α = 0 (no abnormal return). H1: α ≠ 0.	
2. Estimate & SE	intercept of the same P1 regression	α = 0.01335, SE = 0.00741
3. t-value	t = 0.01335 ÷ 0.00741	t = 1.80
4. df / critical / p	df = 58; critical ≈ 2.00; p = 0.077	|1.80| < 2.00
5. Decide	just below the cut-off, p > 0.05	FAIL to reject H0

Meaning: P1's alpha is not significantly different from 0 — consistent with CAPM (no free abnormal return). This is a borderline case (t = 1.80 is close to 2.00): a good reminder that “not significant” isn't “exactly zero,” just “not proven different from zero.”

Appendix A — The exact math the Python code runs

Every number the analysis produces comes from one of the formulas below. The first table lists the simple ones; then we open up the regression “engine” (the matrix maths), and finish with a line-by-line map to the code. Nothing here is beyond what the rest of this guide already prepared you for.

A.1 The simple formulas (data prep and portfolios)

Quantity	Formula	In plain words
Log return	r = ln(P(t) ÷ P(t−1))	this month's price over last month's, in log form — logs add up neatly over time
Risk-free per month	(1 + annual% ÷ 100)^(1/12) − 1	split one year's safe rate into one month's equivalent
Excess return	return − risk-free	reward above the safe asset — everything in CAPM is built on this
Portfolio return	average of its 10 members' returns	an equal-weighted basket
Alpha (α)	mean(y) − β·mean(x)	where the best-fit line crosses zero
Residual	y − (α + β·x)	the gap between the actual point and the line
Residual variance	average of residual²	how scattered those gaps are = firm-specific (unsystematic) risk

A.2 A 60-second matrix primer (only three words)

The regression is written with matrices (grids of numbers). You only need three ideas:
• **Transpose (X^T)** — flip the grid on its side: rows become columns.
• **Multiply (X·y)** — combine two grids by multiplying rows against columns and adding up.
• **Inverse (M^{-1})** — the matrix version of "divide": $M \cdot M^{-1}$ gives the matrix "1". It undoes M, just as 1/5 undoes ×5.
That is all. The famous least-squares recipe **coef = $(X^TX)^{-1}X^T$ y** is simply the multi-input way to find the best-fit intercept and slope — and for a single input it collapses to exactly the friendly formulas you already know: slope = Cov ÷ Var and intercept = mean(y) − slope·mean(x).

A.3 The regression engine: finding the coefficients

The code's **ols()** function finds the intercept and slope(s) that make the total squared error as small as possible. It does this in one matrix step. **X** is the table of inputs with an added first column of 1s (that column is what produces the intercept); **y** is the column of outputs.

Formula

```
coef = (X^T · X)^-1 · X^T · y resid = y − X · coef
```

Simple example

Second pass: X = [column of 1s , the 32 betas]; y = the 32 average returns. One matrix step returns [λ_0, λ_1] = [−0.0085, 0.0204]. For a single input this equals slope = Cov ÷ Var, intercept = mean(y) − slope·mean(x).

In the code

Code lines 61–63: XtXi = inv(X.T@X); coef = XtXi@X.T@y; resid = y − X@coef.

A.4 Error variance and standard error (where the ± comes from)

Once the line is fitted, the leftover scatter tells us how **uncertain** each coefficient is. First the error variance **s^2**: the sum of squared residuals shared out over the **degrees of freedom (n − k)** — not n — because k numbers were "used up" estimating the line. Then each coefficient's **standard error** is the square root of the matching diagonal entry of $s^2 \cdot (X^TX)^{-1}$.

Formula

```
s² = (sum of residual²) ÷ (n − k) SE = √( diagonal of s² · (X^T · X)^-1 )
```

Simple example

Second pass: n = 32, k = 2, so df = 30. Out come SE(λ_1) = 0.00406 and SE(λ_0) = 0.00480 — the ± values behind the t-tests in Part 5.

In the code

Code lines 64–66: n,k = X.shape; s2 = (resid@resid)/(n−k); se = sqrt(diag(s2*XtXi)). Dividing by n−k (not n) is exactly why "degrees of freedom" matters.

A.5 t-value and p-value (turning ± into a decision)

Each coefficient is compared to zero: **t = coefficient ÷ its SE** (how many standard errors from 0). The **p-value** is the tail area of the t-distribution beyond that t, read at **df = n − k**. The code returns t directly; the p-value uses the very same t-distribution that the printed t-table is built from.

Formula

```
t = coef ÷ SE df = n − k p = tail area beyond |t| (t-distribution, df)
```

Simple example

λ_1: t = 0.0204 ÷ 0.00406 = 5.03, df = 30 → p = 0.0000212 (the exact chain shown in Part 5.1).

In the code

Code line 67: t = coef/se. The p-value is obtained from scipy's t-distribution (same idea as the table lookup).

A.6 R-squared and adjusted R-squared (quality of fit)

R^2 compares the scatter the line leaves behind (RSS) to the scatter there was to start with (TSS = how far the y-values sit from their own average). 1 = the line explains everything, 0 = it explains nothing. **Adjusted R^2** then subtracts a penalty for how many inputs were used, so it only rises when a new input truly helps.

Formula

```
R² = 1 − RSS ÷ TSS , RSS = Σ residual² , TSS = Σ (y − mean(y))² adj R² = 1 − (1−R²)(n−1) ÷ (n−k−1)
```

Simple example

Second-pass Model 1: R^2 = 0.458 — beta explains about 46% of the spread in the 32 average returns.

In the code

Code line 68: r2 = 1 − (resid@resid) ÷ ((y − y.mean())²).sum() — literally 1 − RSS/TSS.

A.7 Line-by-line map: code → formula → meaning

Code	Formula	Plain meaning
np.cov(x,y)/np.var(x)	β = Cov(x,y) ÷ Var(x)	beta: how y moves with x (also the regression slope)
XtXi = inv(X.T@X)	(X^T · X)^-1	the “divide” half of least squares
coef = XtXi@X.T@y	(X^T·X)^-1 · X^T·y	best-fit intercept and slope(s)
resid = y − X@coef	y − X·coef	the leftover errors the line misses
s2 = (resid@resid)/(n−k)	RSS ÷ (n − k)	error variance, shared over degrees of freedom
se = sqrt(diag(s2*XtXi))	√ diag(s²·(X^T X)^-1)	± precision (standard error) of each coefficient
t = coef/se	coef ÷ SE	how many SEs the coefficient is from 0

Code	Formula	Plain meaning
r2 = 1 − RSS/TSS	1 − RSS ÷ TSS	share of the variation the model explains

Bottom line: the code never does anything mysterious — it computes exactly these formulas, which are the same ones taught in Parts 2 and 3, just written once in compact matrix form so it can handle all 32 portfolios at once.

One-page cheat sheet

Term	One-line meaning	Where you see it
Mean	Average value	“Average/Mean Return”
Variance / SD	Spread = risk	“Variance of Residuals”
Null H0	“= 0” default	$\lambda = 0$, $\alpha = 0$
α (significance)	Allowed false-alarm rate	* = 5%, ** = 1%
Standard error	± precision of estimate	behind every t-value
t = est ÷ SE	how many SEs from 0	“t-Statistic” columns
df	n − (things estimated)	30 (2nd pass), 58 (1st pass)
critical t	cut-off from t-table	2.042 (df 30), 2.00 (df 58)
p-value	tail area; < 0.05 = reject	“Prob.” columns
Confidence interval	est ± crit·SE; excludes 0 → sig.	range form of the test
R^2 / adj R^2	variation explained (penalised)	below each table
β, λ_0, λ_1	risk sensitivity; intercept & premium	first/second pass

The golden rule. A number is “significant” when $|t| >$ critical t (≈ 2), equivalently $p < 0.05$, equivalently its 95% confidence interval excludes 0 — three ways of saying the same thing. Your real λ_1: $t = 5.03 > 2.042$, $p = 0.00002$, CI excludes 0 → all three agree it is significant.